

Luca Moldovan

Computer Science Student | Software Engineering • AI/ML • Systems

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PROFILE

Third-year BSc Computer Science student at Constructor University with a Mathematics minor and hands-on experience spanning research, software engineering, enterprise systems, databases, and AI automation. Built GPU-backed genomics pipelines, full-stack Python/C/SQL applications, and an award-winning AI research automation agent. Comfortable moving between low-level systems, production-oriented software, and deep-learning workflows; seeking Computer Science roles where strong engineering and analytical problem-solving are valued.

TECHNICAL SKILLS

Programming: Python, C++, C, Rust, Kotlin, SQL, JavaScript, COBOL

AI / Data: PyTorch, Pandas, machine learning, deep learning, LLM/API integration, genomics/bioinformatics

Web / Databases: Flask, SQLite, SQLAlchemy, Jinja2, HTML/CSS, Playwright, relational data modeling

Systems / Dev: Linux, IBM z/OS, IBM Z, JCL, Docker, Slurm, Git/GitHub, CUDA/GPU computing, Qt

EDUCATION

BSc Computer Science, Minor in Mathematics — Constructor University

Sep 2024 - Expected Jun 2027 | Bremen, Germany

- Coursework and academic work across programming, advanced mathematics, computer architecture, databases, operating systems, and software engineering.

Baccalaureate in Mathematics & Informatics — National College "Alexandru Papiu Ilarian"

Sep 2020 - Jun 2024 | Targu Mures, Romania

- Intensive informatics track; recognized in regional Mathematics/Informatics Olympiads and contributed as a programmer for robotics teams.

EXPERIENCE

Research Intern / RISE-Globalink Scholarship Recipient — University of Calgary

Jun 2026 - Present | Calgary, AB, Canada

- Researching genomic representation learning and variant-effect modeling to characterize biology and support disease prediction.
- Implemented exact in-silico mutagenesis (ISM) and gradient-based Taylor ISM (TISM), then designed a hybrid TISM-screening -> ISM-validation workflow for efficient candidate prioritization and concordance testing.
- Built reproducible Python/PyTorch pipelines for variant generation, inference, evaluation, and reporting; executed large GPU jobs on advanced research computing infrastructure using Linux, CUDA, and Slurm.

Teaching Assistant - Elements of Calculus — Constructor University

Feb 2026 - Jun 2026 | Bremen, Germany

- Supported tutorials and office hours, graded coursework, and created supplementary explanations to help students solve calculus problems independently.

IBM Z Student Ambassador / Mainframe Club Lead — IBM Z Student Ambassador Program

Mar 2025 - Jun 2026 | Bremen, Germany

- Led 15+ technical workshops and mentoring sessions on IBM Z and Z Xplore, helping students build enterprise-computing skills and earn IBM badges.
- Presented "Enhancing Skills and Opportunities with IBM Z XPLORE" at the AMC Young Talents Mainframe Convention 2025, hosted at IBM Headquarters in Böblingen, Germany.
- Built and led a campus mainframe community focused on continuous technical learning, collaboration, and exposure to enterprise systems.

Software Development Intern — Living Mainframe GmbH [Project](#)

Jun 2025 | Böblingen, Germany

- Engineered and deployed a mainframe-oriented task-management/integration tool using Flask, Python, C, and SQL/SQLite to address real client feature requirements.
- Integrated a compiled C workflow with a Python/Flask backend, database, and web interface; gained hands-on experience with z/OS, Linux, mainframe architecture, and COBOL/JCL test cases.
- Worked within professional development workflows, team meetings, and client-focused software delivery.

SELECTED PROJECTS

Hackathon Winner - Market Research Automation Agent [GitHub](#)

Python • Playwright • Pandas • SQLite • Gemini API • n8n • Google Sheets

- Built an AI automation pipeline for the Constructor Hackathon Fall 2025 "Automations with AI" track, organized by Constructor Knowledge Labs in collaboration with Constructor Tech.
- Automated competitor discovery, website extraction, structured LinkedIn enrichment, forum research, data cleaning, and master-table consolidation for EdTech/LMS market research.
- Used Gemini API in the agent workflow and prepared structured outputs for n8n orchestration and Google Sheets integration.

Campus Information Center - Database-Driven Web Application [GitHub](#)

Python • Flask • SQL • HTML/CSS • Jinja2

- Developed a full-stack campus information system using normalized relational schemas and ISA hierarchies to model buildings, events, departments, facilities, and personnel.
- Implemented database-backed application logic, queries, user handling/security coursework, and a responsive Flask/Jinja2 interface.

Software Engineering Bug Simulator [GitHub](#)

C++ • Qt • Agile/Scrum • Clean Code

- Contributed to a real-world-style course project using biweekly team rotations, rapid onboarding to unfamiliar repositories, strict sprint deadlines, and structured handovers.
- Focused on modular C++/Qt implementation, maintainability, documentation, architectural clarity, and production-ready code that successor teams could continue immediately.

Self-Hosted Home Lab / Network Attached Storage

Linux • Raspberry Pi 5 • OpenMediaVault • Networking • Security

- Built and administered a self-hosted storage environment with Linux-based services, secure remote access, access controls, and network configuration as a practical systems-administration lab.

AWARDS & RECOGNITION

- 2026 Mitacs Globalink Research Internship Award / DAAD-RISE Worldwide - University of Calgary research placement.
- Constructor Hackathon Fall 2025 Winner - "Automations with AI: Build Agents that Sense, Decide, and Act to Automate Real Tasks."
- Speaker - AMC Young Talents Mainframe Convention 2025 at IBM Headquarters in Böblingen, Germany.

LEADERSHIP & ADDITIONAL EXPERIENCE

- Workshop Coordinator, Erasmus+ "Mindful Living" (Jul 2025) - coordinated five Digital Detox & Self-Management sessions for an international group.
- Workshop Coordinator, Erasmus+ "Young Digitals" (Jul 2024) - led five sessions on digitalization in sports and collaborative web content creation.
- College Representative, Constructor University (Dec 2024 - Jul 2025) - represented 50+ peers and liaised with university staff and residential-life managers.
- BMW Group x Constructor University Hackathon Participant (Nov 2024) - exposure to causal machine learning and automotive failure-prediction workflows.

CERTIFICATIONS & LANGUAGES

Certifications: IBM Z XPLORE - Advanced; IBM Z XPLORE - Concepts; Cloud Computing: Understanding Core Concepts.

Languages: English (native/bilingual), Romanian (native/bilingual), Italian (professional working), German (elementary).